Lei Zhang

zhanglneupsy@outlook.com

Academic Employment

Rotman Research Institute, University of Toronto	Toronto, Canada
Postdoctoral fellow	2023.9 – now
Supervisor	Dr. Claude Alain

Education

McGill University	Montreal, Canada
Joint PhD Training Program	2021.9 – 2022.8
Supervisor	Dr. Robert Zatorre
The University of Chinese Academy of Sciences	Beijing, China
Ph.D., Cognitive neuroscience	2017.9 – 2023.7
Supervisor	Dr. Yi Du
Nanjing Agricultural University	Nanjing, China
BA., Land resources management_JSB EC	2013.9 – 2017.7

Research Interests

Multisensory integration | Episodic memory | Speech perception | Music perception | Cognitive aging | Experience-related (ie. musical training) neuroplasticity

Peer-reviewed Publications

1. **Zhang, L.,** Wang, X., Alain, C., & Du, Y. (2023). Successful aging of musicians: Preservation of sensorimotor regions aids audiovisual speech-in-noise perception. *Science Advances*, 9(17), eadg7056. (**Cover article**, **Top 1% Attention Score** compared to outputs of the same age)
2. **Zhang L**, Ross B, Du Y, Alain C (2025) Long-term musical training can protect against age-related upregulation of neural activity in speech-in-noise perception. *PLOS Biology* 23(7): e3003247. (selected for a **PLOS Biology press release**, covered by more than 100 media outlets, **Top 1% Attention Score**)
3. **Zhang, L.,** Alain, C., & Buchsbaum, B. R. (2026). Different reliance on sensory reinstatement and internally transformed representations during vivid retrieval of visual and auditory episodes. *Journal of Neuroscience,* e1576252026. (selected as **JNeurosci Featured Research**)
4. You, S., **Zhang, L.,** Wu, G., & Du, Y. (2026). The structural dynamics of music drive acute stress recovery through functional reorganization of stress-regulation

networks. *Proceedings of the National Academy of Sciences*, *123*(37), e2611584123. **(Top 1% Attention Score** compared to outputs of the same age)

5. **Zhang, L.,** Binns, M., Chow, R., Rabi, R., Anderson, N. D., Lu, J., ... & Alain, C. (2026). Neural features underlying high-accuracy classification of amnestic mild cognitive impairment using multi-sensory-evoked potentials. GeroScience, 48(2), 2545-2562.
6. **Zhang, L.**, Du, Y., & Zatorre, R. J. (2025). Visual Congruency of Performers' Movements Enhances Vocal Music Reward through Mu Entrainment. *Social Cognitive And Affective Neuroscience*, nsaf089.
7. **Zhang, L.**, & Du, Y. (2022). Lip movements enhance speech representations and effective connectivity in auditory dorsal stream. *NeuroImage*, 119311.
8. **Zhang, L.**, Fu, X., Luo, D., Xing, L., & Du, Y. (2021). Musical experience offsets age-related decline in understanding speech-in-noise: type of training does not matter, working memory is the key. *Ear and Hearing*, 42(2), 258.
9. Jin, X., **Zhang, L.**, Wu, G., Wang, X., & Du, Y. (2024). Compensation or Preservation? Different Roles of Functional Lateralization in Speech Perception of Older Non-musicians and Musicians. *Neuroscience Bulletin*, 1-15.

Preprints

Zhang, L., Yuan, M., Gilboa, A., & Alain, C. (2025). Theta-mediated conceptual reinstatement in vmPFC precedes perceptual reinstatement in ventral visual cortex during memory recall. bioRxiv, 2025-12. (*Journal of Neuroscience,* under 2rd round review)

Manuscripts in prep.

Zhang, L., Yuan, M., Zhang, H., Moscovitch, M., & Alain, C. Learned associations facilitate perceptual decisions through constrained anticipatory visual dynamics and enhanced sensory and motor preparation.

Book chapter

1. **Zhang, L.,** Vuong, V., Bernstein, L., Alain, C., (2025) Effects of Ageing on Musical Emotions and Memory. Oxford Handbook of Music Neuroscience and Neurorehabilitation. (In press)
2. Alain, C., **Zhang, L.,** Vuong, V., Bernstein, L., (2025) Effects of Ageing on Music Perception. Oxford Handbook of Music Neuroscience and Neurorehabilitation. (In press)
3. Vuong, V., **Zhang, L.,** Bernstein, L., Alain, C., (2025) Effects of Ageing on Music Production. Oxford Handbook of Music Neuroscience and Neurorehabilitation. (In press)

Teaching Experience

University of Toronto, Toronto, Canada
Guest Lecturer | 2025–2026

- Course: JNS1000Y (Graduate Course)
- Lecture: Machine Learning in Cognitive Neuroscience
- Teaching evaluation: 7.0/7.0

Conference presentation

1. **Zhang, L.,** Yuan, M., Gilboa, A., & Alain, C. (2026, March). Theta-mediated conceptual reinstatement in vmPFC precedes perceptual reinstatement in ventral visual cortex during memory recall. Poster presentation at CNS 2026
2. **Zhang, L.,** Ross, B., Du Y., & Alain, C. (2024, July). Accumulated reserves hold back age-related neural compensation in speech-in-noise perception. Oral presentation at SMPC 2024
3. **Zhang, L.,** & Alain, C. (2024, March). Auditory Dorsal Stream Connectivity Supports the Older Musicians' Preserved Audiovisual SIN Perception and Visual Benefit. Poster presentation at CNS 2024
4. **Zhang, L.**, & Du, Y., (2020, October). Visual speech cues aid speech motor areas encoding phonemes in noise. Oral presentation at SNL 2020
5. **Zhang, L.**, & Du, Y., (2019, May). Musical training offsets age-related decline of speech in noise perception in perceiving auditory working memory. Oral presentation at international symposium on Language Embodiment: Language Science in the Era of Artificial Intelligence 2019

Academic Service

Editorial Board Membership

Incoming Editorial Board Member, *Communications Psychology* (Nature Portfolio)

Ad-hoc Reviewer

Imaging Neuroscience; Annals of the New York Academy of Sciences; Psychonomic Bulletin & Review; Humanities and Social Sciences Communications; European Journal of Neuroscience; Neuroscience; Brain Topography; Cognitive Neurodynamics; BMC Psychology; Quarterly Journal of Experimental Psychology; Frontiers in Psychology; Frontiers in Neuroscience

Awards

Interdisciplinary Contest In Modeling, Honorable Mention	2016
International symposium on Language Embodiment, poster, first prize	2019
BABRI Doctoral Forum, Excellence Award	2019
Chinese Academy of Science Presidential Scholarship	2023

Mentorship & Supervision

Xiaoqian Zhou, Master's student: Received an excellent grade for thesis and defense.	2023
Mohan Yuan, undergraduate: Accepted into the highly selective Research Specialist Program in Psychology, University of Toronto (10 students per year).	2025

Funding

Joint PhD Training Program Scholarship
Agency: University of Chinese Academy of Sciences
Role: Applicant
Budget: $ 30400
Duration: 2021-2022

The changes in elderly care methods and lifestyles during the urbanization process and their impact on the mental health of the elderly
Agency: University of Chinese Academy of Sciences Innovation Practice Training Program
Role: Leader
Budget: ￥10000
Duration: 2014-2015

The Changes in elderly care and lifestyles during urbanization and their impact on depression in the elderly
Agency: Challenge Cup Project of Nanjing Agricultural University
Role: Leader
Budget: ￥19600
Duration: 2015-2016

Failed funding application

Marie Skłodowska-Curie Actions Postdoctoral Fellowship
How people can remain cognitively-able in old age
Agency: Horizon Europe
Role: Applicant
Duration: 2026-2027
Results: Seal of Excellence

Newton International Fellowship
Agency: the Royal Society
Role: Applicant

Duration: 2026-2027

Canadian Neuroanalytics Scholars Program
Deciphering the Neuroimaging Signatures of Major Depressive Disorder: A CAN-BIND Dataset Analysis for Biomarker Discovery and Treatment Outcome Prediction
Agency: Alberta Neuroscience
Role: Applicant
Duration: 2026-2027

Language & Skills

Languages:	Mandarin (native), English (fluent)
Research techniques:	Multimodal MRI, M/EEG, Psychophysics
Programming language:	Matlab, R, Python